

CloudLabs - 2U | Learner Documentation

Table of Contents

CloudLabs - 2U Learner Documentation.....	1
Table of Contents.....	1
Troubleshooting and FAQs for learners.....	2
FAQs.....	2
Lab VMs Deployment.....	2
How to Launch the Lab.....	2
How to Start/Stop an Instance.....	4
How to Find Environment Details.....	7
How to Switch Between Instances.....	7
How to Delete an Environment.....	8
How to Relaunch an Instance.....	9
How Idle Tracking Works.....	10
Common Troubleshooting.....	11
Support contact.....	14

Troubleshooting and FAQs for learners

This section offers basic troubleshooting steps and FAQs for learners while performing the lab. Having these steps on hand can help them independently resolve common issues.

FAQs

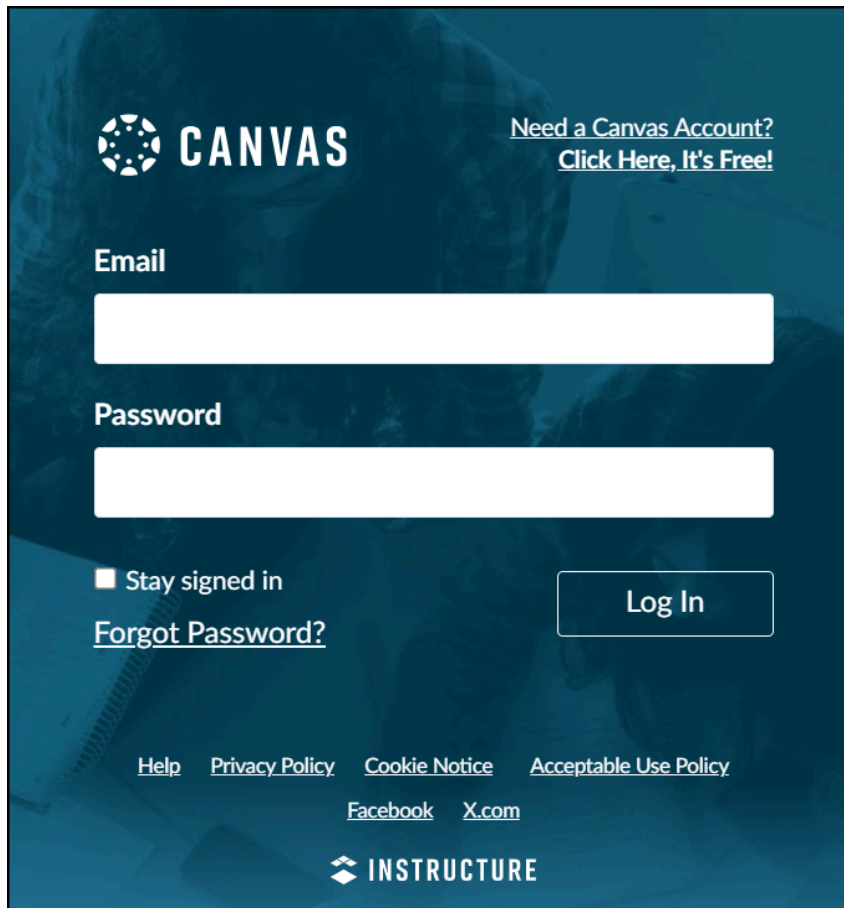
Lab VMs Deployment

By default, **you do not need to deploy any VMs in advance**. CloudLabs platform will automatically deploy VMs when learner launches the lab environment. Learners may have to wait for 5 to 20 minutes, based on labs before the environment is ready.

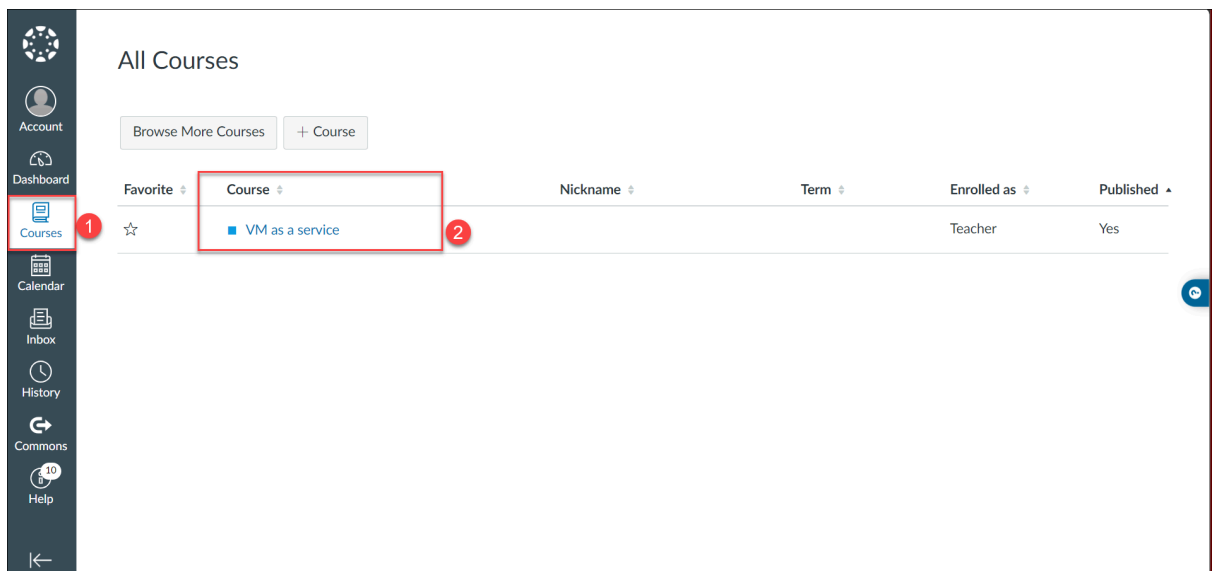
How to Launch the Lab

To launch a lab from the Canvas LMS portal, you'll typically follow these steps. Note that the exact process may vary depending on how your organisation has configured the integration between Canvas and the lab environment, but here's a general guide:

1. Open your web browser and [Log In to Canvas \(instructure.com\)](https://instructure.com)
2. Enter your username and password, then click "**Log In**" to access your Canvas dashboard.



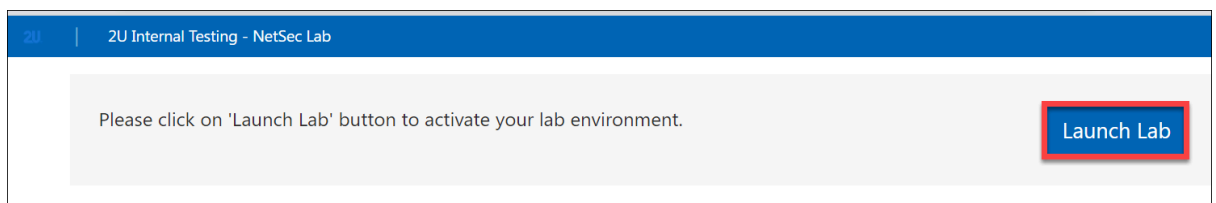
3. From the Canvas dashboard, locate and click on the **course** and select your course.



4. In the course menu on the left side, find and click on the **"Modules"** tab and go to your module where labs are listed. navigate to the specific lab module.



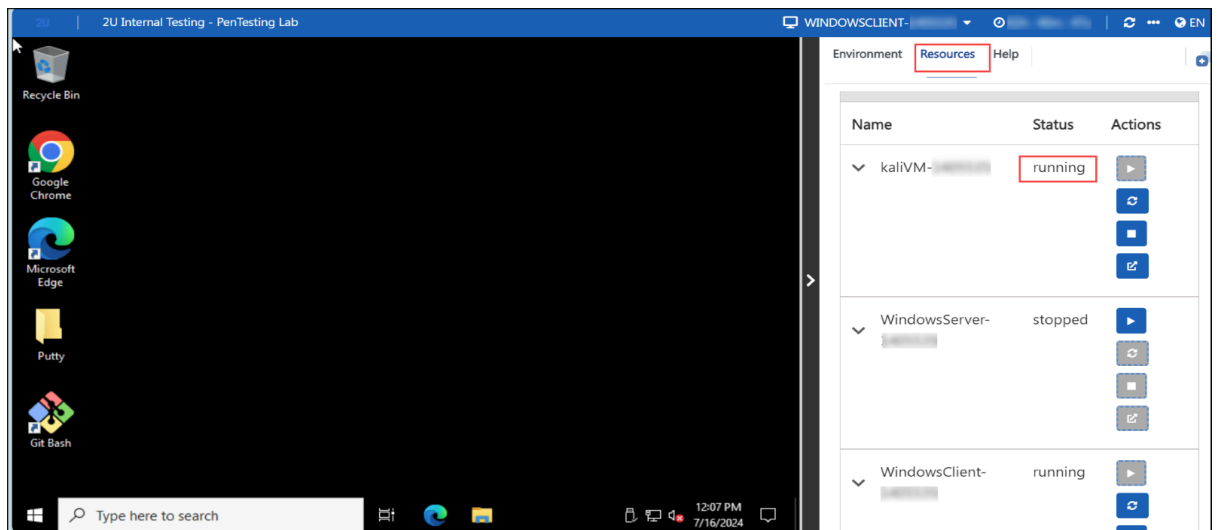
5. Within the module, Select the lab assigned to you . It may be labelled relevant to the module.
6. Read through the provided lab instructions and overview. This will include details about the lab objectives.
7. Click on the assigned lab / Lab Link and it will be navigated to CloudLabs lab launch page.
8. Launch the Lab Environment clicking on button "**Launch Lab**".



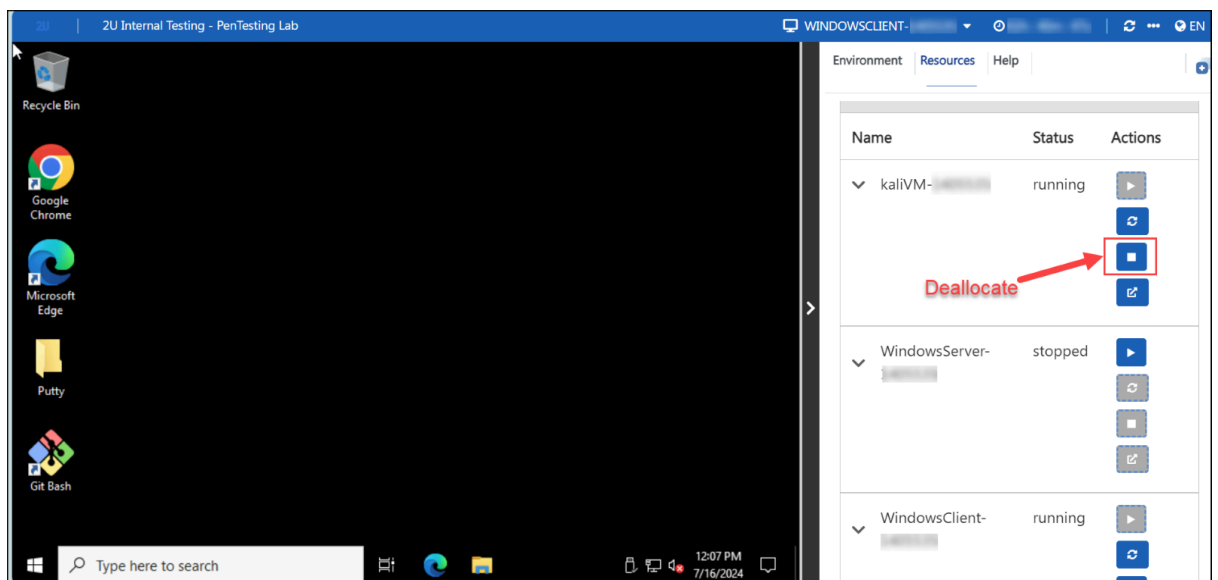
9. After launching the lab, the deployment process will begin. Wait for the deployment to complete, which may take a few minutes. Once done, you will be automatically connected to the virtual machine via RDP over HTTPS.
10. Always access the labs through Canvas or the provided lab link for subsequent visits.

How to Start/Stop an Instance

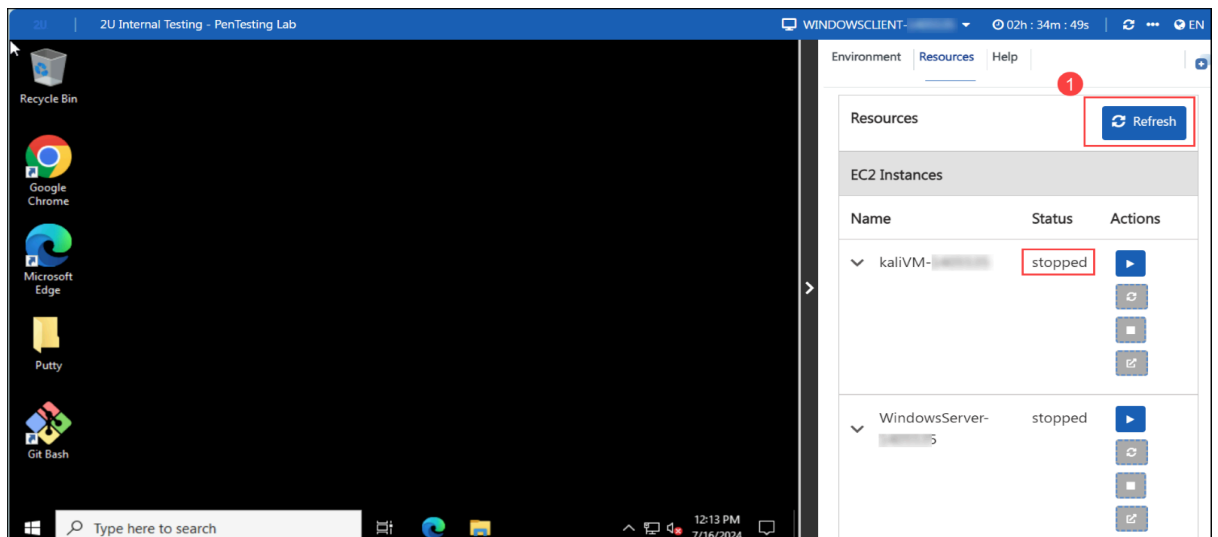
1. Go to the **Resources** tab to view the status of the instances.



2. If you no longer need to use an instance, you can deallocate it by going to the **Resources** tab and following the deallocation steps mentioned below.

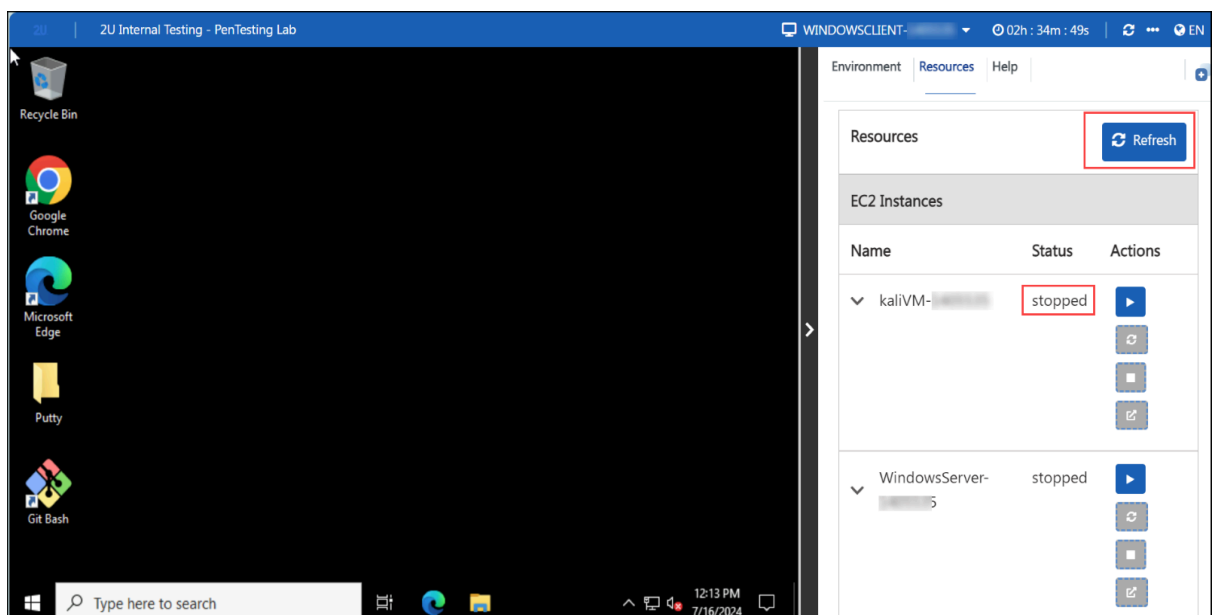


3. Wait for 1-2 minutes, then check the status of the instance to ensure it is in a **stopped** state.

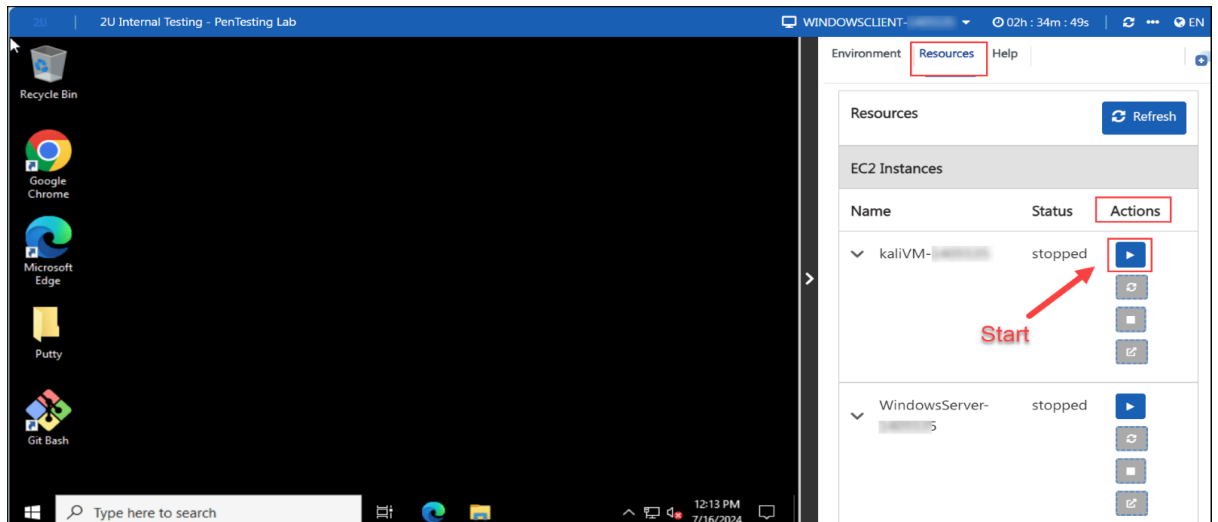


Steps to Start an Instance:

1. Go to the **Resources** tab to view the status of the instances.



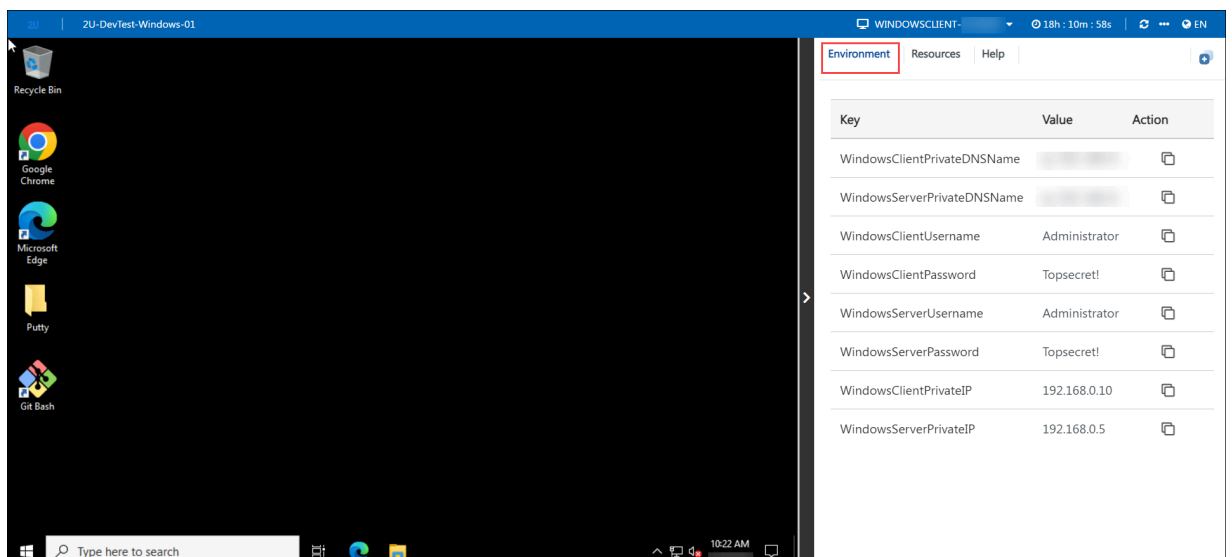
2. Click on **Start** under the **Actions** menu for the instance you want to start.



3. Wait for 1-2 minutes, then check the status of the instance to ensure it is in a running state.

How to Find Environment Details

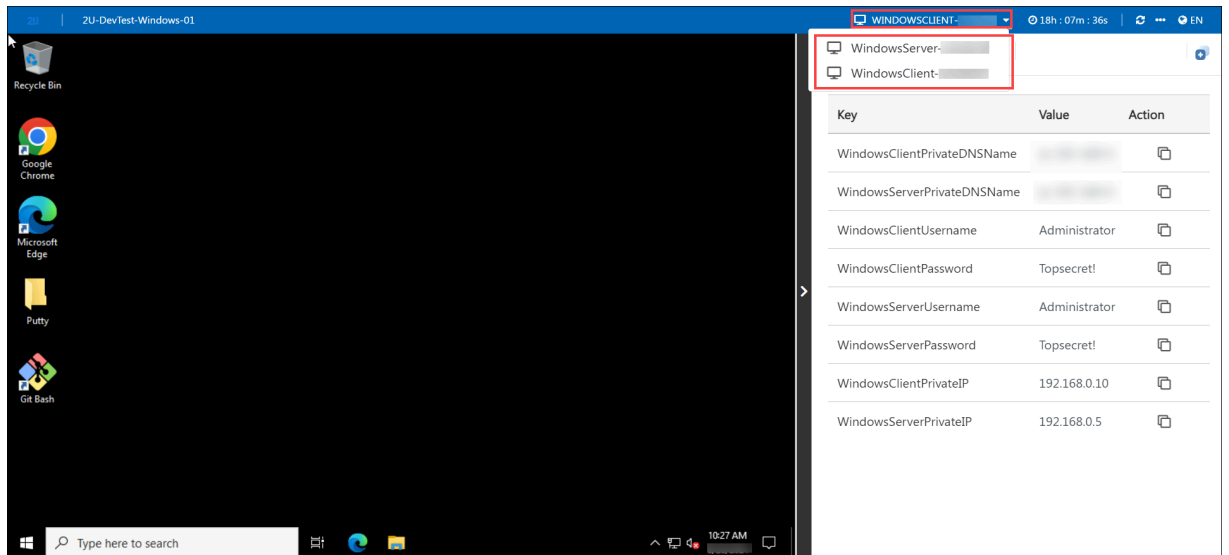
Go to the **Environment** tab to view the lab's environment details.



How to Switch Between Instances

This scenario will come into picture in case of labs with multiple Instances

To switch between instances, go to the top right of the page, next to the duration timer. You'll see the Instance you're currently connected to. Click the dropdown arrow to view a list of instances. Select the instance you want to switch to, and you will be navigated to that instance.

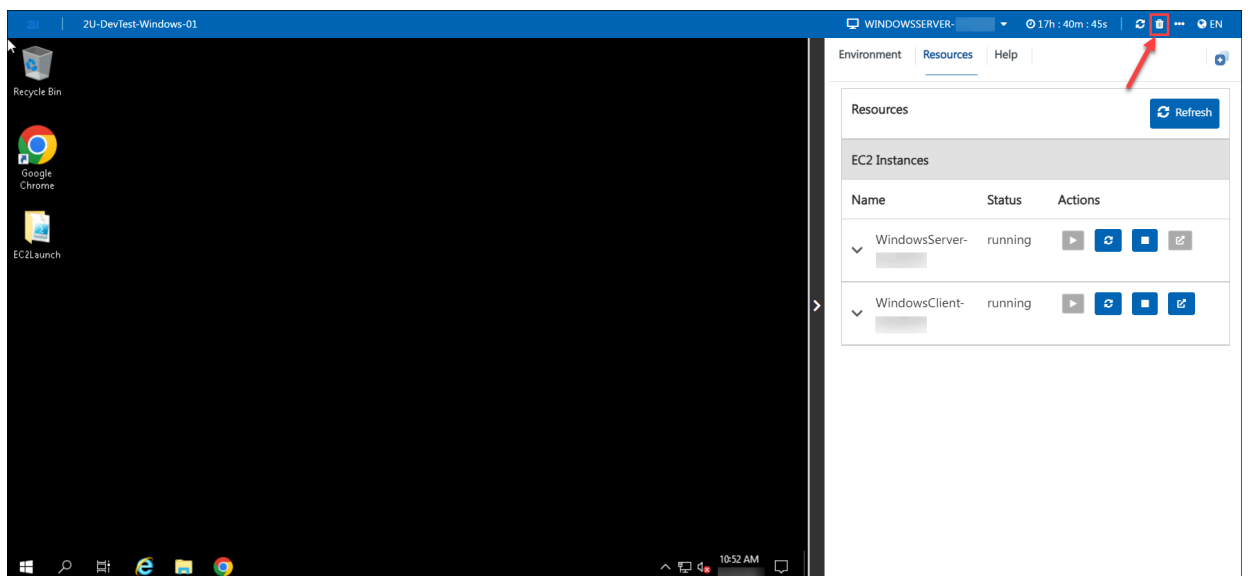


How to Delete an Environment

If you have **completed** your lab testing before the full duration of the lab has elapsed and no longer need access to the lab environment, you can manually delete the environment. This action will ensure that all associated instances are removed from the backend.

You may also choose to delete the environment if you need a **fresh instance** to restart the lab from the beginning for additional testing or practice.

To Delete the environment, navigate to the environment page. In the top right corner, click on the **trash icon** and confirm deletion. This action will completely delete your existing environment, and you will no longer be able to access the lab.



How to Relaunch an Instance

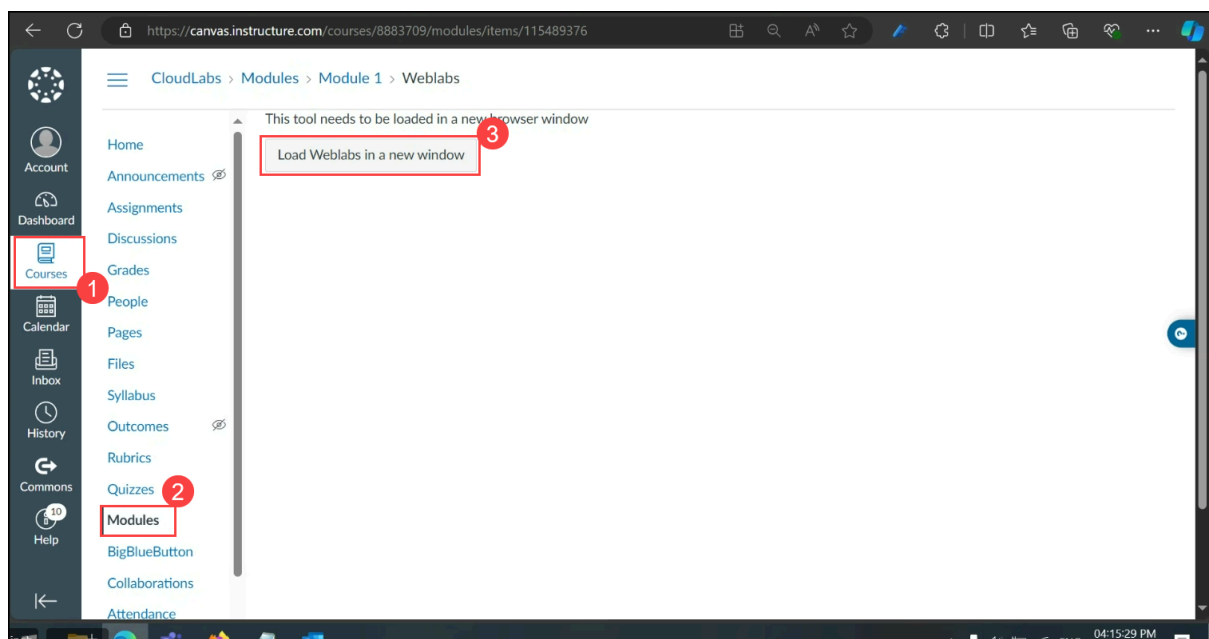
After manually deleting the lab environment as outlined in the previous steps, you can re-access the same lab up to two more times.

To re-access the lab:

1. **Navigate to the Canvas Module:** Go back to the specific module in Canvas where you initially deployed the lab that you just deleted.
2. **Initiate the Lab Deployment:**
 - Locate the option to "**Load in a New Tab**" within the module.
 - Click this option to initiate the deployment of a new lab environment.
3. **Access the Fresh Instance:**
 - A new lab environment will be deployed, providing you with fresh instances.
 - You can then access these instances and proceed with the lab steps as before.

Note: You can only reaccess the same lab environment two more times after the initial deployment.

Please refer to the attached screenshot for a visual guide on how to re-access the lab in Canvas.



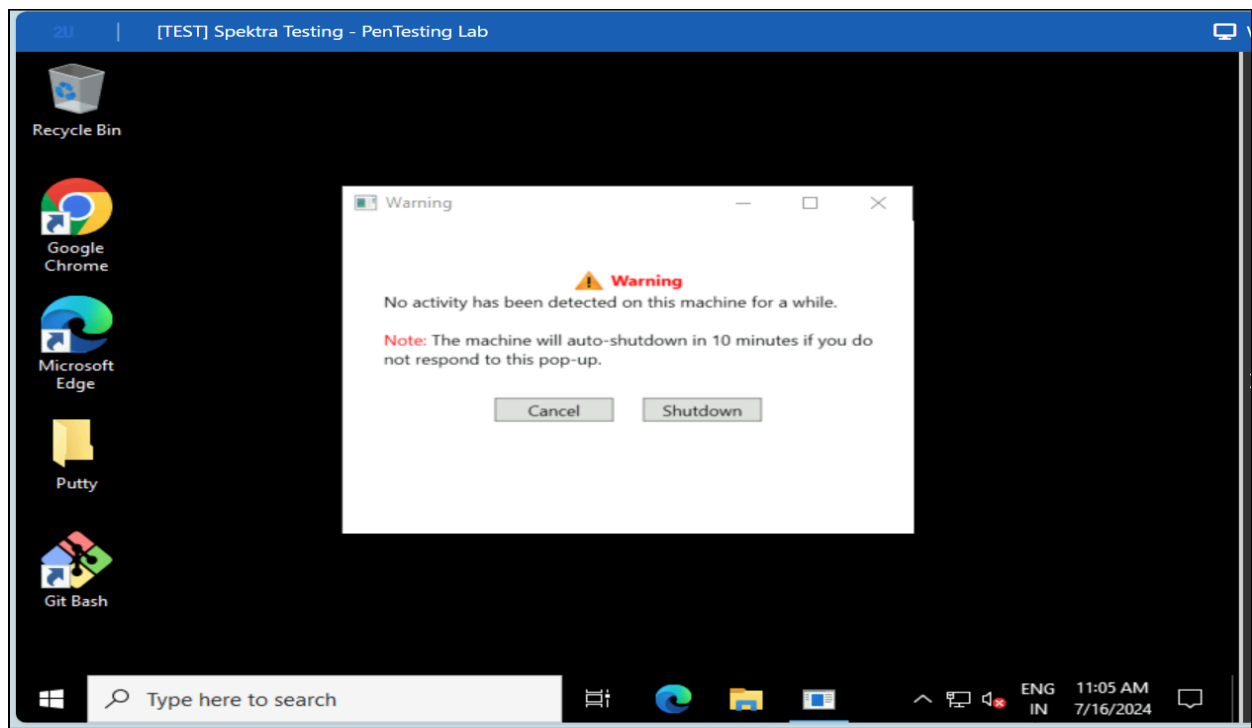
How Idle Tracking Works

We have set up idleness tracking for the instances that are part of the lab environment. If an instance remains idle (i.e., no tasks are performed inside the instance) for 35 minutes, a

popup will appear stating that no activity has been detected and the instance will automatically shut down after 10 minutes.

- **Popup Notification:**
 - After 35 minutes of inactivity, a popup window will appear, informing the user that no activity has been detected and the instance will auto-shutdown in 10 minutes.
- **User Actions:**
 - **Cancel Button:** If the user clicks on the **Cancel** button, a fresh timer will start, and idleness detection will begin again.
 - **Shutdown Button:** If the user clicks on the **Shutdown** button, the instance will immediately shut down.

Refer to the screenshot below for the popup window that users will see after 25 minutes of inactivity.

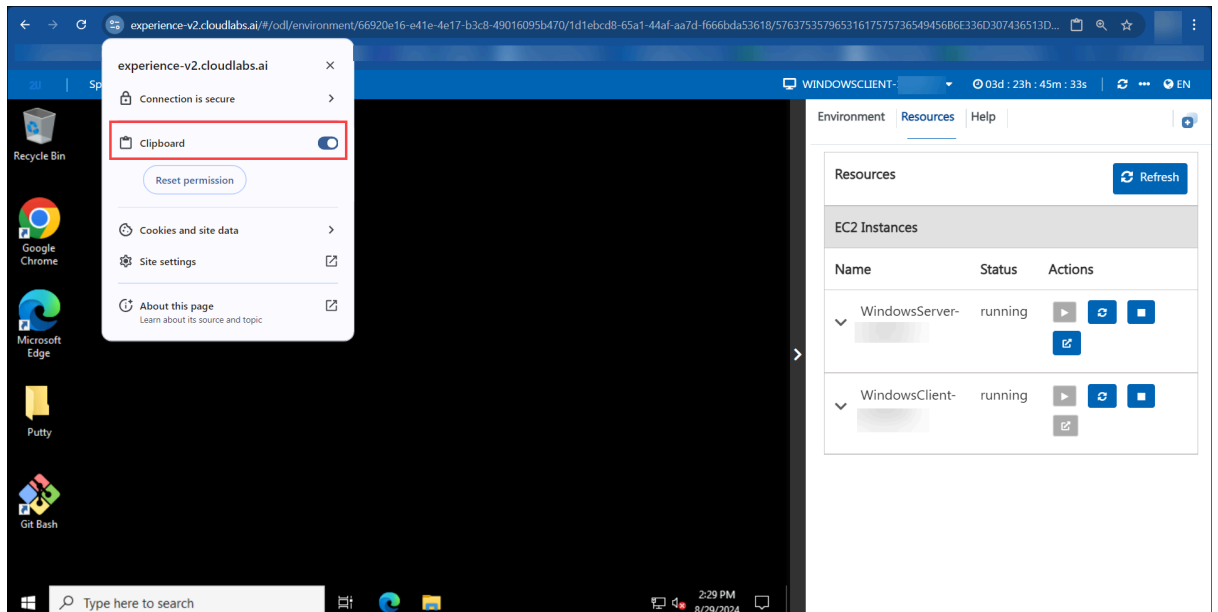


Common Troubleshooting

Unable to Copy/Paste in lab environment

If Copy/Paste functionality doesn't work within your lab environment, try the following steps:

- Please check if Clipboard permissions are on **Enabled**, if not then enable it.



- After setting it to Allow, select the content which you want to copy from the lab guide, then right click > Copy to copy the content and then paste it on the desired location in the Lab VM.
- Try to refresh the page or try any other Web Browser.
- Check your Network bandwidth in the local System/Laptop.

RDP: Known Functionality Issues

This section covers some of the common issues while using your labs. Sometimes, JumpBox/LabVM RDP Connection (RDP Gateway connection) won't work at attendee's network due to the following reasons:

Common Issues

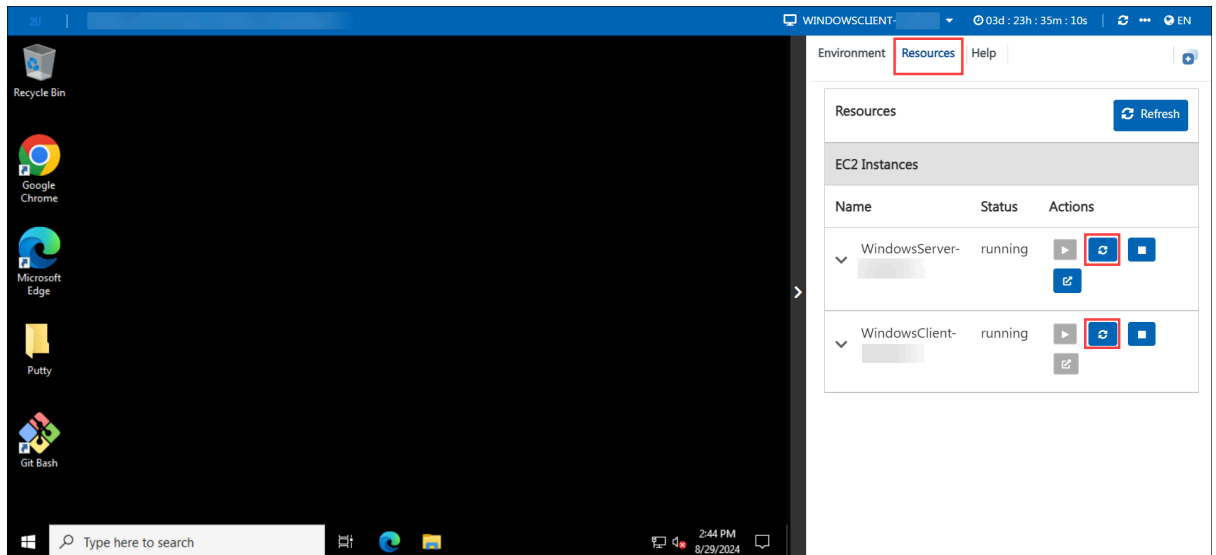
Here are some of the most common scenarios due to which users may face problems using/accessing the lab and its many features.

- Organisation firewall/policy blocking the connection.
- User is connected to VPN (Virtual Private Network), which may restrict the connection based on network policies.
- Issues with cookies and cache in the internet browser.
- Low network bandwidth (Network bandwidth recommended: 8 Mbps).
- Https Over RDP access denied due to **“Websocket connection blocked”**.

Troubleshooting Checklist

These are some of the basic checklists before you proceed with other troubleshooting guides/support.

- Make sure you are connected to a good internet connection with minimum network bandwidth of 8 Mbps.
- Clear the cache from the browser, Enable Cookies.
- Launch the lab in **Private/Incognito** browsing mode.
- Restart the VM from the Lab Environment Page.



Fix "Websocket connection blocked" to access HTTPS over RDP

Possible reasons

- AdBlocker / Cookie blocker browser extensions
- Antivirus and Firewall software
- Proxy and VPN connections

Solutions

1. If you are already working in Google Chrome, try the **Guest mode**. If you still get the error even after using **Guest mode** of your browser, your antivirus or Firewall may be blocking it.
2. Temporarily turn off all antivirus and Firewall programs or try a different device (for example an iPad).
3. If the error is fixed, your antivirus or Firewall is probably the cause.
4. If you still get the error, it may be caused by proxy or VPN connections.
5. In such a situation, we can't determine which specific antivirus or firewall software is blocking the websocket connections, or the steps to prevent this from happening. Please contact your IT manager.
6. If you still get the error, it may be caused by proxy or VPN connections.

Important: Please ask your IT manager to allow websockets via **wss://*.ms-cloudlabs.ai**

Enable Cookies in the Browsers

Safari Users

1. Click the Safari menu from the top toolbar.
2. Choose **Preferences**.
3. Click the **Privacy** tab.
4. Click the **Never** checkbox for Block Cookies.

Chrome Users

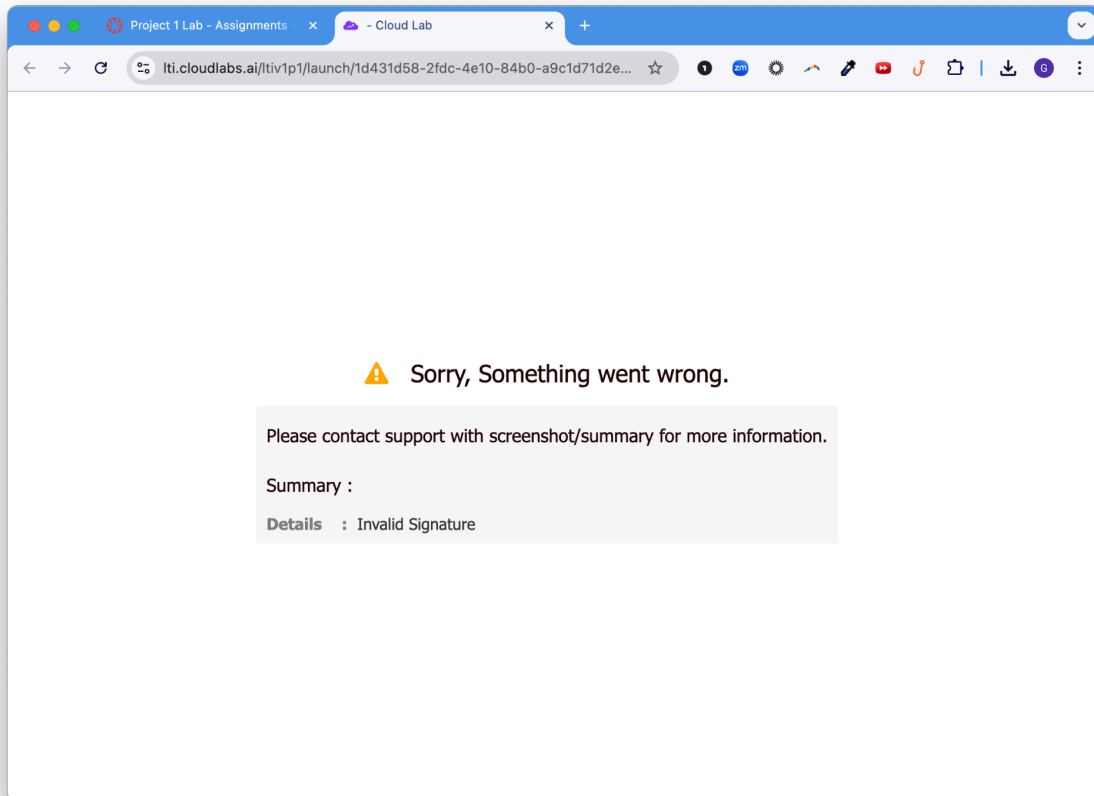
1. Click the context menu in the browser toolbar to the right of the address bar.
2. At the top right, click **More** and then **Settings**.
3. Under "Privacy and security," click Site settings.
4. Click **Cookies**.
5. From here, you can:
 - Turn on cookies: Next to "Blocked," turn on the switch.
 - Turn off cookies: Turn off Allow sites to save and read cookie data.

Firefox Users

In the Menu bar at the top of the screen, click Firefox and select Preferences.

1. Click the menu button and select Options.
2. Select the Privacy & Security panel.
3. To allow cookies, make sure that the Checkbox for "Cookies" is unchecked.
4. Close the page. Any changes you've made will automatically be saved.

Invalid Signature Error



Possible Solutions

- Clear cache and cookies in the browser, sign in to the LMS again and try re-launching the lab
- Use an incognito tab if using Chrome
- Switch to another browser

Support contact

For any lab issues or questions, please contact the respective instructors.